

IIB43203 ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme	:	Bachelor In Internet Of Things
Course Code & Name	:	IIB43203 - CLOUD COMPUTING
Assessment	:	GROUP PROJECT
Dateline	:	Day <u>07</u> Month <u>06</u> Year <u>2026</u>
Penalties	:	• 10% will be deducted for late submission. • Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly. • AI content will affect your mark

Section B: Academic Integrity

Tick (✓) each box below if you agree:

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☒	This submission is my own, unless indicated with proper referencing.
☒	This submission has not been previously submitted or published.
☒	This submission follows the requirements stated in the course.

Section C: Submission Detail

Office Receipt of Submission

No	Student Name(s)	Student ID(s)
1	AMEER AFIQ BIN ARJUNA	52227124345
2	AHMAD FARHAN BIN RIZMAL	52227124355
3	MUHAMMAD FAZIL BIN FIHIR	52227124253
4	ISMAIRIL BIN MUHAMMAD RIZAL	52227124581

Student Receipt of Submission

The submission receipt of the student task will be based on the **VLE submission date and time**.



MALAYSIAN INSTITUTE OF INFORMATION TECHNOLOGY

MARCH 2026 SEMESTER

CLOUD COMPUTING (IIB43203)

GROUP PROJECT

PREPARED BY:

NO	NAME	STUDENT ID NO.	HP NO.
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EVALUATE BY:

Dr. Megat Norulazmi

SUBMISSION DATE:

07/06/2026

Instruction Guide

Create VPC.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☐ VPC only

☒ VPC and more

Name tag auto-generation [Info](#)

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

cc-project

IPv4 CIDR block [Info](#)

Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16

65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

- Click Create VPC.
- Select VPC and more (This creates subnets, route tables, and gateways automatically).
- Name tag auto-generation: cc_group_project
- IPv4 CIDR block: 10.0.0.0/16

Number of Availability Zones (AZs) [Info](#)

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 | **2** | 3

► Customize AZs

Number of public subnets [Info](#)

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 | **2**

Number of private subnets [Info](#)

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 | **2** | 4

► Customize subnets CIDR blocks

NAT gateways (\$) - *updated* [Info](#)

NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.

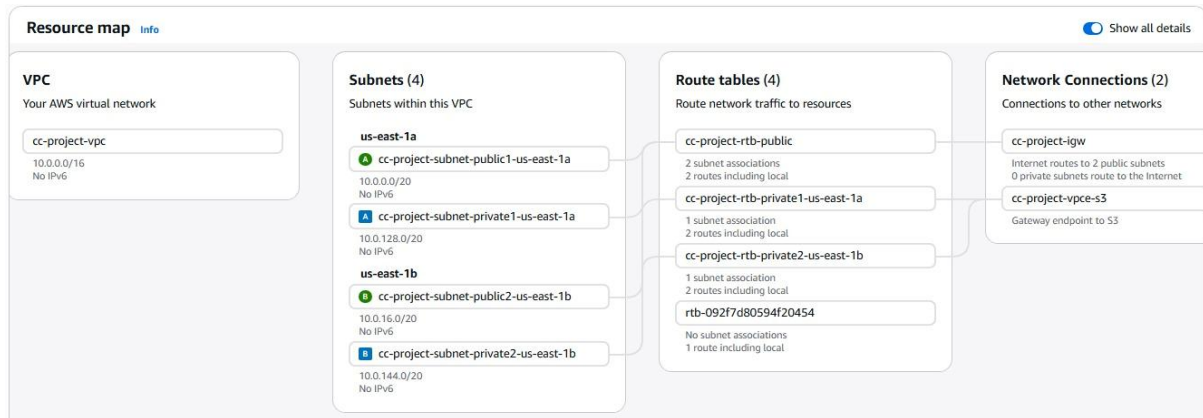
None | Regional - new | Zonal

VPC endpoints [Info](#)

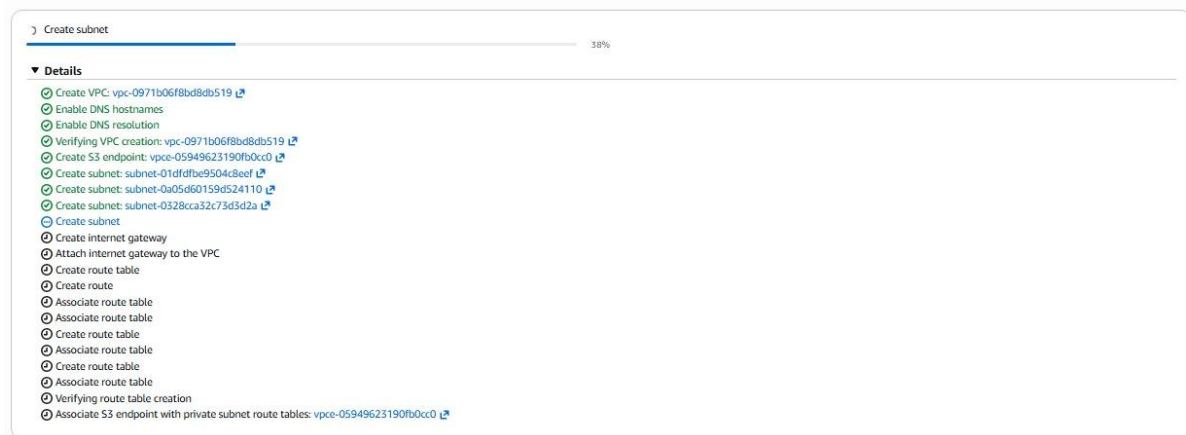
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None | **S3 Gateway**

- Select VPC and more (This creates subnets, route tables, and gateways automatically).
- Number of Availability Zones (AZs): Select 2 (for high availability).
- Number of public subnets: 2
- Number of private subnets: 2
- NAT gateways: Select In 1 AZ (This is cheaper for a student project).



Create VPC workflow



[VPC](#) > [Your VPCs](#) > vpc-0971b06f8bd8db519 ⓘ 🔍

vpc-0971b06f8bd8db519 / cc-project-vpc Actions

Details [Info](#)

VPC ID vpc-0971b06f8bd8db519	State ✔ Available	Block Public Access ⊖ Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0851aff34bd85cd12	Main route table rtb-092f7d80594f20454
Main network ACL acl-0774ac77ff48da09e	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 491893219727
Encryption control ID -	Encryption control mode -		

- You should now see 1 VPC, 2 Public Subnets, 2 Private Subnets, 1 Internet Gateway (for public access), and 1 NAT Gateway (for private internet access)

Create the Web Server Security Group

Create security group

Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

Info

cc_project_web_sg

Name cannot be edited after creation.

Description

Info

Security group for web server - allows HTTP and SSH

VPC

Info

vpc-0971b06f8bd8db519 (cc-project-vpc)

- Security group name: cc_group_project-web-sg
- Description: Security group for web server - allows HTTP and SSH
- VPC: Select your cc_group_project VPC (should start with "vpc-xxxxx")

Inbound rules

Info

Type

Info

HTTP

Protocol

Info

TCP

Port range

Info

80

Source

Info

Anyw...

Description - optional

Info

Allow web traffic from internet

Delete

SSH

TCP

22

My IP

0.0.0.0/0

115.135.28.25/32

Allow SSH access only from my computer

Delete

Add rule

Outbound rules

Info

Type

Info

All traffic

Protocol

Info

All

Port range

Info

All

Destination

Info

Custom

0.0.0.0/0

Description - optional

Info

Delete

Add rule

Click Add rule and add these two rules:

Rule 1 - HTTP (for website visitors):

- Type: HTTP
- Protocol: TCP (auto-filled)
- Port range: 80
- Source: Anywhere-IPv4 (0.0.0.0/0)
- Description: Allow web traffic from internet

sg-03278f559f974ae89 - cc_project_web_sg

Actions

Details

Security group name

cc_project_web_sg

Owner

491893219727

Security group ID

sg-03278f559f974ae89

Inbound rules count

2 Permission entries

Description

Security group for web server - allows HTTP and SSH

Outbound rules count

1 Permission entry

VPC ID

vpc-0971b06f8bd8db519

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	
☐	-	sgr-048851b7ff3f313b0	IPv4	HTTP	TCP	80	0.0.0.0/0	AI
☐	-	sgr-0a085ccb36d02aa2b	IPv4	SSH	TCP	22	115.135.28.25/32	AI

Rule 2 - SSH (for you to manage the server):

- Type: SSH
- Protocol: TCP (auto-filled)
- Port range: 22
- Source: My IP (This automatically adds your current IP address)
- Description: Allow SSH access only from my computer

Outbound Rules (Leave as default)

Default allows all outbound traffic (your server can reach the internet)

Create the Database Security Group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

cc_project_db_sg

Name cannot be edited after creation.

Description [Info](#)

Security group for database - only accepts traffic from web server

VPC [Info](#)

vpc-0971b06f8bd8db519 (cc-project-vpc)

Configure Basic Settings

- Security group name: cc_group_project-db-sg
- Description: Security group for database - only accepts traffic from web server
- VPC: Same cc_group_project VPC

Inbound rules [Info](#)

Type [Info](#)

MySQL/Aurora

Protocol [Info](#)

TCP

Port range [Info](#)

3306

Source [Info](#)

Custom

Q sg-03278f559f974ae8 X

sg-03278f559f974ae89 X

Description - optional [Info](#)

Allow MySQL traffic only from web servers

Allow MySQL traffic only from web servers

Delete

Add rule

Outbound rules [Info](#)

Type [Info](#)

All traffic

Protocol [Info](#)

All

Port range [Info](#)

All

Destination [Info](#)

Custom

Q

0.0.0.0/0 X

Description - optional [Info](#)

Delete

Add rule

Outbound Rules (Leave as default)

Default allows all outbound traffic

sg-0bc6d43625b82b056 - cc_project_db_sg

[Actions](#)

Details

Security group name
cc_project_db_sg

Security group ID
sg-0bc6d43625b82b056

Description
Security group for database - only accepts traffic from web server

VPC ID
[vpc-0971b06f8bd8db519](#)

Owner
491893219727

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Inbound rules (1)

[Manage tags](#) [Edit inbound rules](#)

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0e64693056a7ad1f5	-	MySQL/Aurora	TCP	3306	sg-03278f559f974ae89...	Allow

Add Inbound Rule (Only ONE rule needed)

- Type: MySQL/Aurora (or search for it in the dropdown)
- Protocol: TCP (auto-filled)
- Port range: 3306
- Source: Select "Custom" from dropdown, then type/select: cc_group_project-web-sg
- Description: Allow MySQL traffic only from web servers

Security Note: This is crucial! The source is the OTHER security group, NOT an IP address. This means only resources using the cc_group_project-web-sg can access your database. Nothing from the internet can reach it directly.

Create a Database Subnet Group (Tells RDS which subnets to use)

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

cc_project-db-subnet-group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

Private subnets for cc_group_project database

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

cc-project-vpc (vpc-0971b06f8bd8db519)
4 Subnets, 2 Availability Zones

Configure Subnet Group

- Name: cc_group_project-db-subnet-group
- Description: Private subnets for cc_group_project database
- VPC: Select your cc_group_project VPC (starts with vpc-xxxxx)

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a

us-east-1b

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

cc-project-subnet-private1-us-east-1a

Subnet ID: subnet-0328cca32c73d3d2a CIDR: 10.0.128.0/20

cc-project-subnet-private2-us-east-1b

Subnet ID: subnet-0188fecde31ed3e75 CIDR: 10.0.144.0/20

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	cc-project-subnet-private1-us-east-1a	subnet-0328cca32c73d3d2a	10.0.128.0/20
us-east-1b	cc-project-subnet-private2-us-east-1b	subnet-0188fecde31ed3e75	10.0.144.0/20

Add Subnets

- Under "Availability Zones", select 2 zones (us-east-1a, us-east-1b - or whatever your region shows)
 - Under "Subnets", ONLY select the PRIVATE subnets from your VPC
 - Look for subnets with names containing "private" or "private-us-east-1a" and "private-us-east-1b"
- DO NOT select public subnets (this keeps your database secure)

Create the Database Instance

Choose a database creation method

Full configuration

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

Easy create

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates

Choose a sample template to meet your use case.

Production

Use defaults for high availability and fast, consistent performance.

Dev/Test

This instance is intended for development use outside of a production environment.

Free tier

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Create database [Info](#)

Free plan has access to limited features and resources

The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

[Upgrade plan](#)

Engine options

Engine type [Info](#)

☒ Aurora (MySQL Compatible)



☐ Aurora (PostgreSQL Compatible)



☒ MySQL



☐ PostgreSQL



☐ MariaDB



☐ Oracle

ORACLE

☐ Microsoft SQL Server

Microsoft SQL Server

☐ IBM Db2

IBM Db2

Select Standard Create (not "Easy Create")

- Engine type: MySQL
- Version: MySQL 8.0.xx (default is fine)

Settings

Edition

☒ MySQL Community

Engine version [Info](#)

View the engine versions that support the following database features.

► Show filters

Engine version

MySQL 8.4.8 ▼

☐ Enable RDS Extended Support [Info](#)

Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

cc-project-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*

☒ Self managed

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength [Info](#) Very strong

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

Settings

- DB instance identifier: cc-group-project-db
- Master username: admin
- Master password: Create a strong password like CcGroup2024! (WRITE THIS DOWN!)
- Confirm password: Same password

Connectivity

- VPC: cc_group_project (your VPC)
- DB subnet group: cc_group_project-db-subnet-group (should be auto-selected)
- Public access: Select NO (This keeps database off the internet)
- VPC security group: Choose "Choose existing"
- Existing VPC security groups: SELECT cc_group_project-db-sg
- Availability Zone: No preference (or select one)

▼ Additional configurations

Connectivity & security

Endpoint & port Endpoint  cc-project-db.cwb6gsswwzx6.us-east-1.rds.amazonaws.com Port 3306	Networking Availability Zone us-east-1b VPC cc-project-vpc (vpc-0971b06f8bd8db519) Subnet group cc_project-db-subnet-group Subnets subnet-0328cca32c73d3d2a subnet-0188fecde31ed3e75 Network type IPv4	Security VPC security groups cc_project_db_sg (sg-0bc6d43625b82b056)  Active Publicly accessible Yes Certificate authority Info rds-ca-rsa2048-g1 Certificate authority date May 26, 2061, 07:34 (UTC+08:00) DB instance certificate expiration date June 07, 2027, 05:18 (UTC+08:00)
--	---	---

Find Your Database Endpoint

- Click on your database name: cc-group-project-db
- Look in the Connectivity & security tab
- COPY THIS ENDPOINT and save it in a text file. You'll need it in Step 5.
- Creating Your Web Server with EC2

Launch the EC2 Instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

- Name Your Server
- Name: cc_group_project-web-server

▼ Application and OS Images (Amazon Machine Image) [info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-00e801948462f718a (64-bit (x86), uefi-preferred) / ami-0b183bb1259186479 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Choose Amazon Machine Image (AMI)

- Scroll to "Application and OS Images"
- Select Amazon Linux 2023 AMI (Free tier eligible)
- Architecture: 64-bit (x86)

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-00e801948462f718a

Publish Date

2026-05-21

Username

ec2-user



Verified provider

▼ Instance type [info](#) | [Get advice](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

Choose Instance Type

- Select t2.micro (Free tier eligible)
- choose t3.micro

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

cc_project-key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel

Create key pair

Key Pair (Login)

- Click Create new key pair
- Key pair name: cc_group_project-key
- Key pair type: RSA
- Private key file format: .pem (for Mac/Linux) OR .ppk (for Windows)
- Click Create key pair - it will automatically download

SAVE THIS FILE SAFELY - you'll need it if you ever need to SSH into your server

▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-0971b06f8bd8db519 (cc-project-vpc)
10.0.0.0/16



Subnet | [Info](#)

subnet-01dfdfbe9504c8eef cc-project-subnet-public1-us-east-1a
VPC: vpc-0971b06f8bd8db519 Owner: 491893219727 Availability Zone: us-east-1a (use1-az1)
Zone type: Availability Zone IP addresses available: 4091 CIDR: 10.0.0.0/20

[Create new subnet](#)

Auto-assign public IP | [Info](#)

Enable

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups | [Info](#)

Select security groups

[Compare security group rules](#)

cc_project_web_sg sg-03278f559f974ae89
VPC: vpc-0971b06f8bd8db519

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Network Settings (Click Edit)

- VPC: cc_group_project (your VPC)
- Subnet: Choose ANY Public Subnet (e.g., cc_group_project-public-us-east-1a)
- Auto-assign public IP: Enable
- Firewall (security groups): Select existing security group
- Common security groups: SELECT cc_group_project-web-sg

Important: Make sure you select your cc_group_project-web-sg, NOT the default security group

User data - *optional* | [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
# Update system
dnf update -y

# Install web server, PHP, and database tools
dnf install -y httpd php php-mysql mariadb105 wget unzip

# Start web server
systemctl start httpd
systemctl enable httpd

# Download sample PHP application
cd /var/www/html
wget https://github.com/vijaythapa333/php-crud-api/archive/refs/heads/main.zip
unzip main.zip
```

☐ User data has already been base64 encoded

Advanced Details (Scroll down - CRITICAL STEP!)

Find User data section

```
#!/bin/bash
```

```
# Update system
```

```
dnf update -y
```

```
# Install web server, PHP, and database tools
```

```
dnf install -y httpd php php-mysqli mariadb105 wget unzip
```

```
# Start web server
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
# Download sample PHP application
```

```
cd /var/www/html
```

```
wget https://github.com/vijaythapa333/php-crud-api/archive/refs/heads/main.zip
```

```
unzip main.zip
```

```
cp -r php-crud-api-main/* .
```

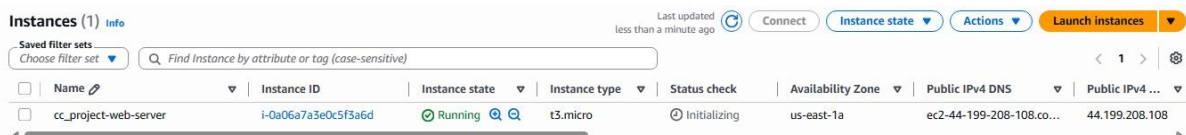
```
rm -rf php-crud-api-main main.zip
```

```
# Set permissions
```

```
chown -R apache:apache /var/www/html
```

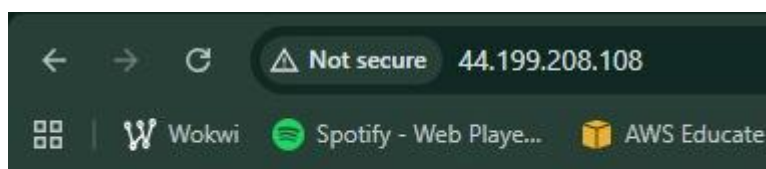
```
chmod -R 755 /var/www/html
```

What this script does: Installs Apache web server, PHP, downloads a sample app, and configures it automatically.



Instances (1) Info							
Last updated less than a minute ago Refresh Connect Instance state Actions Launch instances							
Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)							
☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	cc_project-web-server	i-0a06a7a3e0c5f3a6d	Running	t3.micro	Initializing	us-east-1a	ec2-44-199-208-108.co...

Test Your Web Server (Before Database Connection)



It works!

Find your server's public IP

- In EC2 > Instances, click on cc_group_project-web-serve
- Copy the Public IPv4 address
- Open a web browser and paste

A PHP error message about database connection (this is EXPECTED!)
 Something like: "Connection failed: php_network_getaddresses: getaddrinfo failed"
 This means your web server is **RUNNING** but can't find the database yet

```
PS C:\Users\Ameer Afiq\Downloads> ssh -i cc_project-key.pem ec2-user@44.199.208.108
The authenticity of host '44.199.208.108 (44.199.208.108)' can't be established.
ED25519 key fingerprint is SHA256:F3Xl5hBRuZ9nTXsdkfBbVcZ6GdBd8QRxoPGnFvnlcQw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.199.208.108' (ED25519) to the list of known hosts.

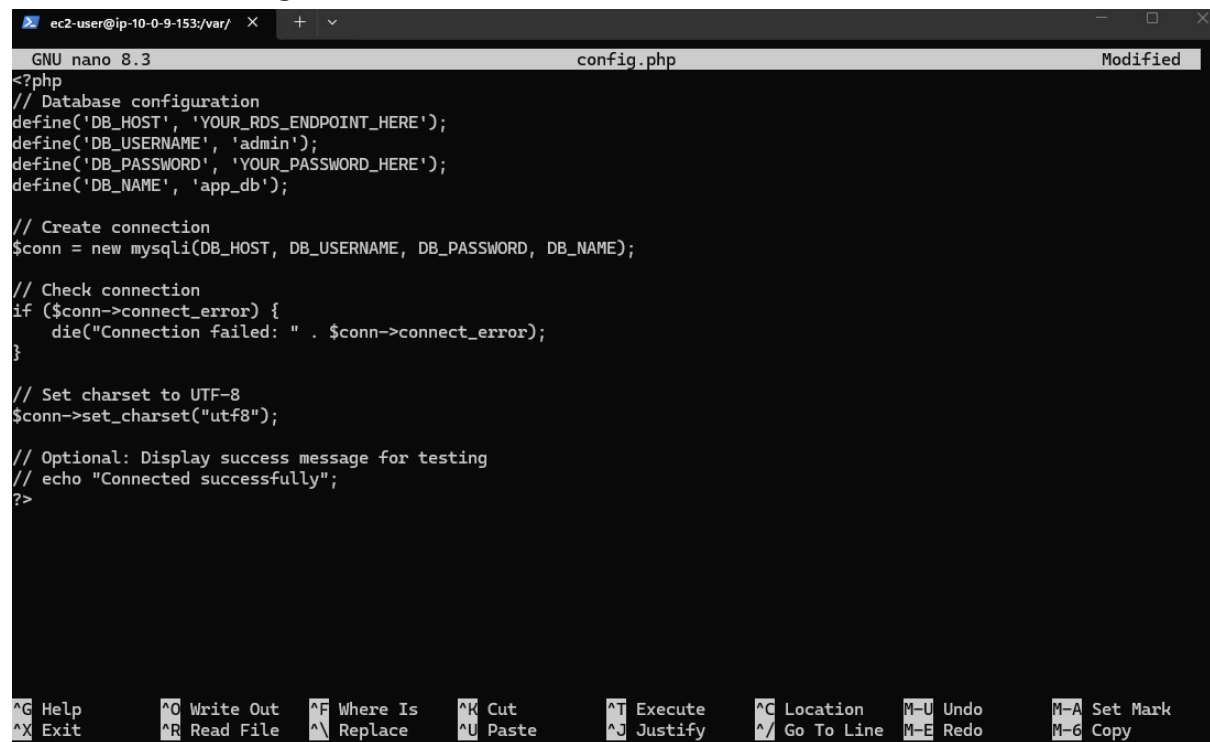
  _
 ~\  #####_      Amazon Linux 2023
  NN  \_#####\
  NN   \###|
  NN   \#/  _--
  NN    V~'  ' ->      https://aws.amazon.com/linux/amazon-linux-2023
  NN
  NN
  NN  .-.-
  NN  /  /
  NN  /m/'

[ec2-user@ip-10-0-9-153 ~]$
```

1. Navigate to the folder containing .pem key file
2. Run the ssh command:
`ssh -i "cc-project-key.pem" ec2-user@<Public-IP>`
3. Type yes when asked to confirm
4. Successful login will show the image above

Run the following command:

Edit database configuration file



```
ec2-user@ip-10-0-9-153:/var/ X + -
GNU nano 8.3 config.php Modified
<?php
// Database configuration
define('DB_HOST', 'YOUR_RDS_ENDPOINT_HERE');
define('DB_USERNAME', 'admin');
define('DB_PASSWORD', 'YOUR_PASSWORD_HERE');
define('DB_NAME', 'app_db');

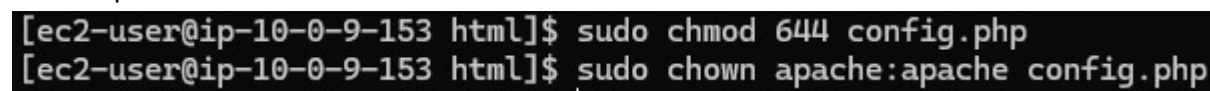
// Create connection
$conn = new mysqli(DB_HOST, DB_USERNAME, DB_PASSWORD, DB_NAME);

// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Set charset to UTF-8
$conn->set_charset("utf8");

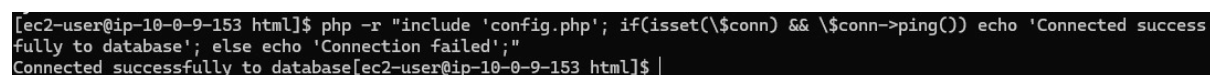
// Optional: Display success message for testing
// echo "Connected successfully";
?>
```

Set file permission



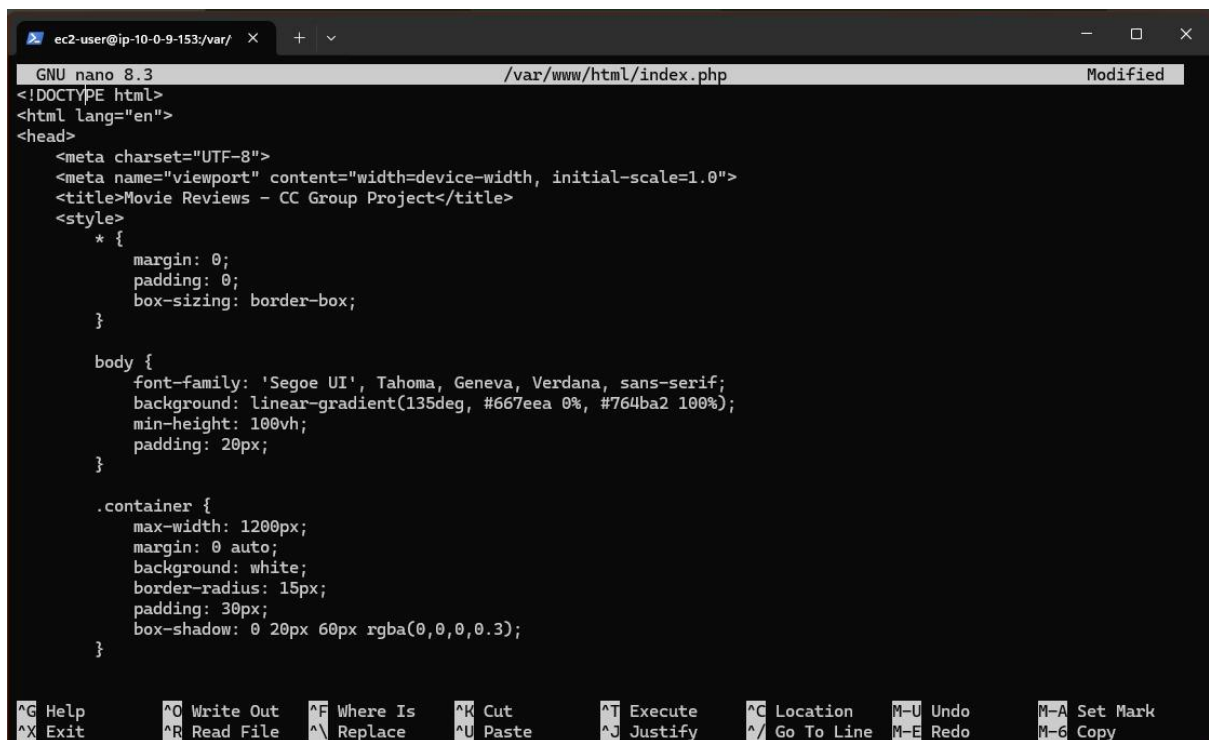
```
[ec2-user@ip-10-0-9-153 html]$ sudo chmod 644 config.php
[ec2-user@ip-10-0-9-153 html]$ sudo chown apache:apache config.php
```

Test database connection



```
[ec2-user@ip-10-0-9-153 html]$ php -r "include 'config.php'; if(isset(\$conn) && \$conn->ping()) echo 'Connected successfully to database'; else echo 'Connection failed';"
Connected successfully to database[ec2-user@ip-10-0-9-153 html]$
```

Edit main website file



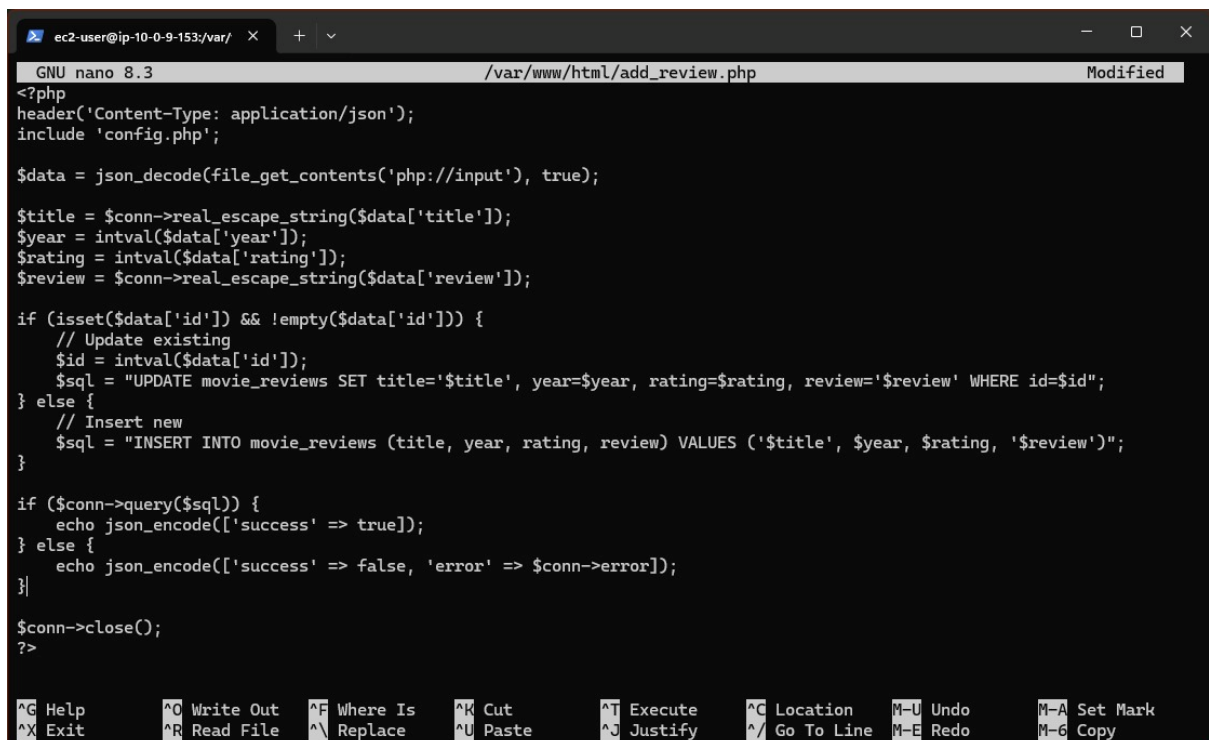
```
GNU nano 8.3 /var/www/html/index.php Modified
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Movie Reviews - CC Group Project</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
      min-height: 100vh;
      padding: 20px;
    }

    .container {
      max-width: 1200px;
      margin: 0 auto;
      background: white;
      border-radius: 15px;
      padding: 30px;
      box-shadow: 0 20px 60px rgba(0,0,0,0.3);
    }
  </style>
</head>
<body>
</body>
</html>
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/_ Go To Line M-E Redo M-6 Copy

Edit review processing file



```
GNU nano 8.3 /var/www/html/add_review.php Modified
<?php
header('Content-Type: application/json');
include 'config.php';

$data = json_decode(file_get_contents('php://input'), true);

$title = $conn->real_escape_string($data['title']);
$year = intval($data['year']);
$rating = intval($data['rating']);
$review = $conn->real_escape_string($data['review']);

if (isset($data['id']) && !empty($data['id'])) {
  // Update existing
  $id = intval($data['id']);
  $sql = "UPDATE movie_reviews SET title='$title', year=$year, rating=$rating, review='$review' WHERE id=$id";
} else {
  // Insert new
  $sql = "INSERT INTO movie_reviews (title, year, rating, review) VALUES ('$title', $year, $rating, '$review')";
}

if ($conn->query($sql)) {
  echo json_encode(['success' => true]);
} else {
  echo json_encode(['success' => false, 'error' => $conn->error]);
}

$conn->close();
?>
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/_ Go To Line M-E Redo M-6 Copy

```
ec2-user@ip-10-0-9-153:/var/ x + v
GNU nano 8.3 /var/www/html/delete_review.php Modified
<?php
header('Content-Type: application/json');
include 'config.php';

$data = json_decode(file_get_contents('php://input'), true);
$id = intval($data['id']);

$sql = "DELETE FROM movie_reviews WHERE id=$id";

if ($conn->query($sql)) {
    echo json_encode(['success' => true]);
} else {
    echo json_encode(['success' => false, 'error' => $conn->error]);
}

$conn->close();
?>
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy

Configure web application permission

```
[ec2-user@ip-10-0-9-153 html]$ sudo chown -R apache:apache /var/www/html
[ec2-user@ip-10-0-9-153 html]$ sudo chmod -R 755 /var/www/html
[ec2-user@ip-10-0-9-153 html]$ sudo systemctl restart httpd
[ec2-user@ip-10-0-9-153 html]$
```

Create Amazon S3 Bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region

US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace

Choose the namespace where you want to create your bucket. [Learn more](#)

☒ Global namespace

By default, S3 creates general purpose buckets in the global namespace.

☐ Account Regional namespace (recommended)

General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

cc-project-bucket

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

- Open the AWS Management Console and create Bucket.
- Enter a unique name and select AWS region

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

- Keep ACLS Disable and leave Block Public Access enabled for security
- Create Bucket

Create IAM roles for EC2 access to Amazon S3

Step 1

☒ **Select trusted entity**

☐ Add permissions

☐ Name, review, and create

Select trusted entity [Info](#)

☒ **AWS service**

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

- Navigate IAM roles and create roles
- Under Trusted Entity Type, select AWS Service
- Choose EC2 under use case and click next

Configure Roles Details

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

cc_project_role_2026

Maximum 64 characters. Use alphanumeric and "+, -, @, _" characters.

Description

Add a short explanation for this role.

Allows EC2 to read/write to S3 buckets for movie posters

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _ + , - @ _ ' # % * ^ & ~ =

- Role name: cc_project_role_2026

- Description: Allows EC2 to read/write to S3 buckets for movie posters

Assign S3 Permissions

Step 2: Add permissions

[Edit](#)

Permissions policy summary

Policy name 	Type	Attached as
AmazonS3FullAccess	AWS managed	Permissions policy
AmazonS3ReadOnlyAccess	AWS managed	Permissions policy

- Search for AmazonS3FullAccess and select it.
- Click next

Verify role creation

Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID
 i-0a06a7a3e0c5f3a6d (cc_project-web-server)

IAM role

Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

cc_project_role_2026

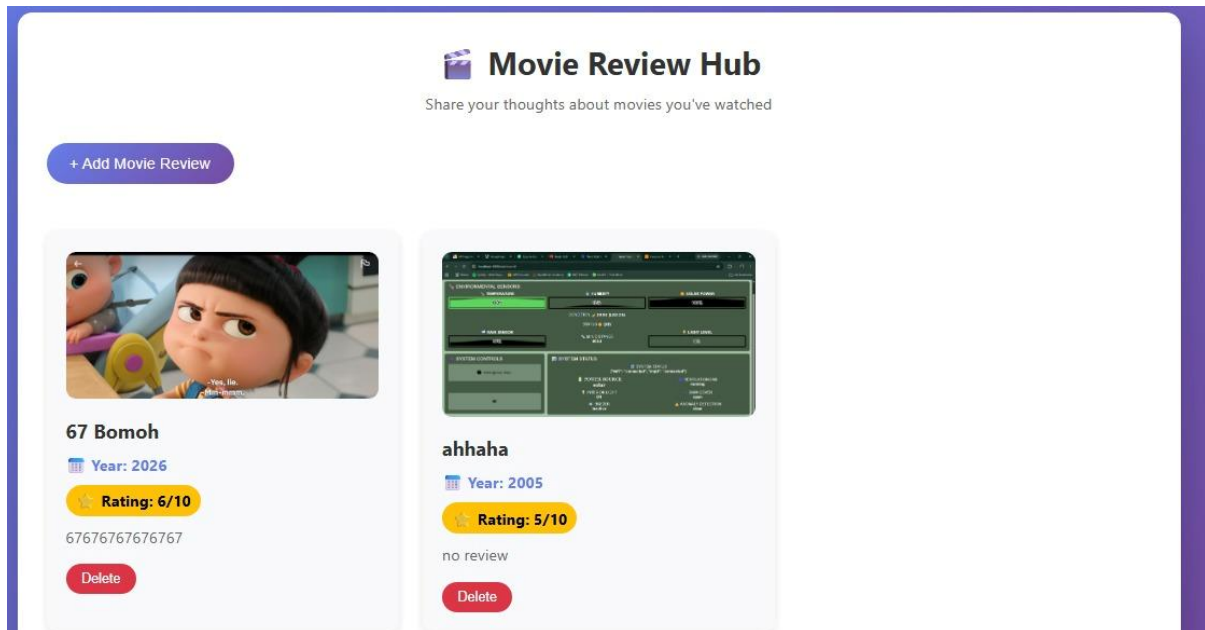
[Create new IAM role](#)[Cancel](#)[Update IAM role](#)

- Return to IAM and search for cc_project_role_2026 and confirm that role has been created successfully and S3 policy is already attached

Output:

This our application that we created called “Movie Review Hub”. This application allow user to browse movie information, write review and give rating to the movies.

Website interface



Page for user to review a movies

Add Movie Review

Movie Title

Release Year

Rating (1-10)

Your review...

Movie Poster (Image):

Choose File

No file chosen

Upload JPG, PNG or GIF (max 5MB)

Submit Review

Cancel

Final Bucket S3

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

Objects (8)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

	Name	Type	Last modified	Size	Storage class
☐	1780826363_Screenshot2026-04-02131631.png	png	June 7, 2026, 17:59:24 (UTC+08:00)	778.4 KB	Standard
☐	1780826366_Screenshot2026-04-02131631.png	png	June 7, 2026, 17:59:28 (UTC+08:00)	778.4 KB	Standard
☐	1780826367_Screenshot2026-04-02131631.png	png	June 7, 2026, 17:59:30 (UTC+08:00)	778.4 KB	Standard
☐	1780827441_Screenshot2026-04-02131631.png	png	June 7, 2026, 18:17:22 (UTC+08:00)	778.4 KB	Standard
☐	1780827759_Screenshot2026-06-07011554.png	png	June 7, 2026, 18:22:40 (UTC+08:00)	97.8 KB	Standard
☐	1780828426_Screenshot2026-04-02131631.png	png	June 7, 2026, 18:33:48 (UTC+08:00)	778.4 KB	Standard
☐	1780828427_Screenshot2026-04-02131631.png	png	June 7, 2026, 18:33:48 (UTC+08:00)	778.4 KB	Standard
☐	1780828649_UniKL.jpg	jpg	June 7, 2026, 18:37:31 (UTC+08:00)	17.6 KB	Standard